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Problem 1

Period	A	B	C
0	–3400	–3400	–3400
1	0	1870	1020
2	0	1870	1700
3	6460	1870	3284

Project	Payback (yrs)	IRR	NPV @10%	NPV @15%
A	2.53	23.86%	1453.49	847.55
B	1.82	29.92%	1250.41	869.63
C	2.21	28.01%	1399.55	931.68

Project A

$$\text{Payback} = 2 + \frac{3400}{6460} = 2.53 \text{ yrs}$$

$$0 = -3400 + \frac{6460}{(1+\text{IRR})^3} \quad \Rightarrow \quad \text{IRR} = 23.86\%$$

$$\text{NPV}_{10\%} = -3400 + \frac{6460}{(1.1)^3} = 1453.49$$

$$\text{NPV}_{15\%} = -3400 + \frac{6460}{(1.15)^3} = 847.55$$

Project B

$$\text{Payback} = 1 + \frac{1530}{1870} = 1.82 \text{ yrs}$$

$$0 = -3400 + \frac{1870}{(1+\text{IRR})} + \frac{1870}{(1+\text{IRR})^2} + \frac{1870}{(1+\text{IRR})^3} \quad \Rightarrow \quad \text{IRR} = 29.92\%$$

$$\text{NPV}_{10\%} = -3400 + \frac{1870}{(1.1)} + \frac{1870}{(1.1)^2} + \frac{1870}{(1.1)^3} = 1250.41$$

$$\text{NPV}_{15\%} = -3400 + \frac{1870}{(1.15)} + \frac{1870}{(1.15)^2} + \frac{1870}{(1.15)^3} = 869.63$$

Project C

$$\text{Payback} = 2 + \frac{680}{3284} = 2.21 \text{ yrs}$$

$$0 = -3400 + \frac{1020}{(1+\text{IRR})} + \frac{1700}{(1+\text{IRR})^2} + \frac{3284}{(1+\text{IRR})^3} \quad \Rightarrow \quad \text{IRR} = 28.01\%$$

$$\text{NPV}_{10\%} = -3400 + \frac{1020}{(1.1)} + \frac{1700}{(1.1)^2} + \frac{3284}{(1.1)^3} = 1399.55$$

$$\text{NPV}_{15\%} = -3400 + \frac{1020}{(1.15)} + \frac{1700}{(1.15)^2} + \frac{3284}{(1.15)^3} = 931.68$$

Problem 2

Period	A	B	A - B
0	–400	–200	–200
1	250	140	110
2	300	179	121

Project	IRR	NPV @9%
A	23.32%	81.86

B	35.87%	79.1
A - B	10%	2.76

$$\text{NPV}_\text{A} = -400 + \frac{250}{(1.09)} + \frac{300}{(1.09)^2} = 81.86$$

$$0 = -400 + \frac{250}{(1+\text{IRR})} + \frac{300}{(1+\text{IRR})^2} \quad \Rightarrow \quad \text{IRR} = 23.32\%$$

$$\text{NPV}_\text{B} = -200 + \frac{140}{(1.09)} + \frac{179}{(1.09)^2} = 79.1$$

$$0 = -200 + \frac{140}{(1+\text{IRR})} + \frac{179}{(1+\text{IRR})^2} \quad \Rightarrow \quad \text{IRR} = 35.87\%$$

$$\text{NPV}_{\text{A - B}} = -200 + \frac{110}{(1.09)} + \frac{121}{(1.09)^2} = 2.76$$

$$0 = -200 + \frac{110}{(1+\text{IRR})} + \frac{121}{(1+\text{IRR})^2} \quad \Rightarrow \quad \text{IRR} = 10\%$$

Decision: Project A.

- B has a greater IRR (35.87% vs. 23.32%)
- Difference in scale makes IRR ranking incorrect
- At 9%, A has a higher NPV (81.86 vs. 79.1).
- Incremental A - B (invest 200 more, get 110 and 121)
  - IRR = 10% > 9%, so the extra investment creates value (incremental NPV = 2.76)
- The NPV profiles cross at 10%
  - Below rate, A is preferred
  - Above rate, B is preferred

Problem 3

T 0

- Old assets are sold under book value
- Loss = 300,000 - 180,000 = 120,000
- Tax shield = 120,000 × 30% = 36,000

T 1-5

- Pre-tax benefit = 200,000 + 50,000 = 250,000
- 175,000 after tax
- Incremental depreciation = 160,000 - 100,000 in years 1-3 (the old assets' depreciation is lost)

$$\text{Years 1-3} : (160{,}000 - 100{,}000) \times 30\% = 18{,}000$$

$$\text{Years 4-5} : 160{,}000 \times 30\% = 48{,}000$$

Period	**0**	**1**	**2**	**3**	**4**	**5**
New equipment	-800,000					
Sale of old assets	180,000					
Tax shield on loss on sale	36,000					
Working capital	-30,000					30,000
After-tax operating benefit		175,000	175,000	175,000	175,000	175,000
Depreciation tax shield		18,000	18,000	18,000	48,000	48,000
Net cash flow	**-614,000**	**193,000**	**193,000**	**193,000**	**223,000**	**253,000**
Discount factor @15%	1	0.8696	0.7561	0.6575	0.5718	0.4972
PV	**-614,000**	**167,826**	**145,936**	**126,901**	**127,501**	**125,786**
NPV						**79,949**

NPV @15% = 79,949.

- The NPV is positive, so the investment earns more than the 15% required return:
- ABC Manufacturing should replace the old equipment

Problem 4

The $50K preliminary study is excluded (sunk cost)

Question 1: NPV of change #1 only

- Book value of the machines = 1000 - 400 = 600
- Selling at 300 is a loss of 300, with a tax shield of 300 × 40% = 120
- Additionally, selling cancels 200/yr of depreciation, i.e. a tax shield of 80/yr in years 1-3

Period	**0**	**1**	**2**	**3**
Sale of machines	300			
Tax shield on loss on sale	120			
Lost depreciation tax shield		–80	–80	–80
Net cash flow	**420**	**–80**	**–80**	**–80**
Discount factor @20%	1	0.8333	0.6944	0.5787
PV	**420**	**–66.7**	**–55.6**	**–46.3**
NPV				**251.5**

NPV of change #1 @20% = 251.5K

Question 2: Relevant cash flows

- Overtime savings: + 10% a year: 100, 110, 121 (without tax)
- Inventory is working capital (recovered at cost in year 3)
- Holding costs are not taken into account

Period	**Time 0**	**End of Year 1**	**End of Year 2**	**End of Year 3**
Sale of machines	300			
Tax shield on loss on sale	120			
Lost depreciation tax shield		–80	–80	–80
Inventory (working capital)	–150			150
Software purchase	–300			
Software depreciation tax shield		40	40	40
After-tax overtime savings		60	66	72.6
Net cash flow	**–30**	**20**	**26**	**182.6**

Question 3: Decision

$$\mathrm{NPV}_{20\%} = -30 + \frac{20}{1.2} + \frac{26}{1.2^2} + \frac{182.6}{1.2^3} = 110.4 \text{ K}$$

The NPV is positive, so C2L should use level production